



UA EcoCAR DriveAbility Website

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General Motivation

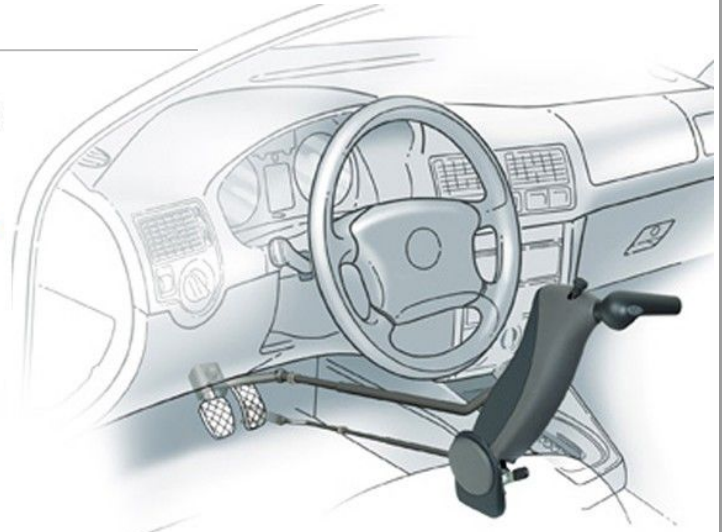
The **DriveAbility** website serves as source of information for adaptive automotive controls. These controls allow people with disabilities to successfully and comfortably drive vehicles.

The website will provide reviews with respect to the compatibility and physical requirements of various hand control models.

**Classic
Hand Control**



**Compact
Hand Control**



Users

The website's user base revolves around the community surrounding adapted driving features. Those who have purchased hand controls are able to share their experience and provide advice to those who are purchasing hand controls.

The data collected by the website can also be used by healthcare professionals to assist in providing tailored recommendations to those in need.



Core Features

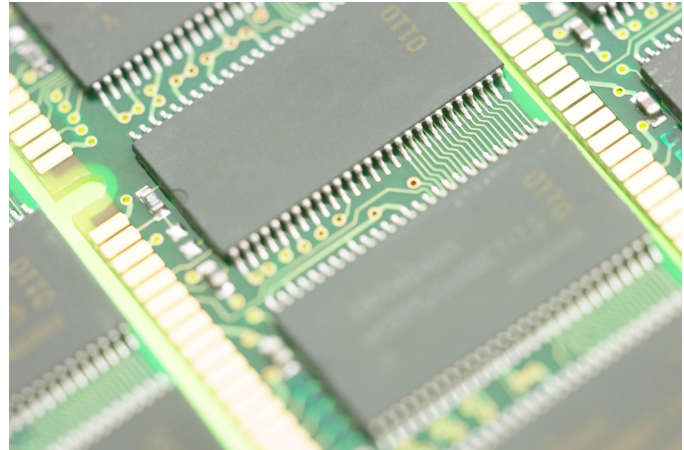
- User Profiles split into Reviewers and Admin
 - Admin approve both reviews and new products
 - Reviewers can view products, read & write reviews, suggest additional products, and mark favorites
- Product Pages containing:
 - Product name, image of product, basic manufacturer info
 - Where to purchase and typical cost
 - Star ratings in multiple categories (comfort, durability, ease of use, value, installation, overall)
 - Experimental grip strength
- Ability to post reviews on products with each review including:
 - Reviewer age, height range, weight range
 - Text box where reviewer writes their comments
 - Star ratings in multiple categories
- Ability to Search for products based on various metrics (name, price, rating, etc...)

Technical Details

The preexisting stack is as follows:

- **Frontend:** Next.js, React, TypeScript, Tailwind CSS
- **Backend:** Node.js, Express.js, TypeScript
- **Database:** MongoDB
- **Authentication:** JWT (JSON Web Tokens)
- **Rich Text Editor:** Quill.js
- **3D Visualization:** Three.js (for ErgoRig model)
- **Deployment:** Docker, Nginx

Focus will be more on expansion of backend functionality



Initial Thoughts on Development



Most of our work will be related to the backend logic of the website, as the front end is close to completion.



We will be developing the login logic, database reading, writing and storage, and how all of the individual pieces of the website will work interactively.

Related Apps



National Mobility Equipment Dealers Association

Nonprofit trade association representing Automotive Mobility Industry. Contains guides, resources, and dealer information for those looking for information on automotive mobility. <https://www.nmeda.org>



GoodReads

A site that allows for reviews of individual books, allowing a user to look up a book and see other users' reviews. We may use a similar model, with user reviews for a specific product congregated together and with a similar rating system, allowing reviewers to comment on specific aspects of the product. <https://www.goodreads.com>



Amazon

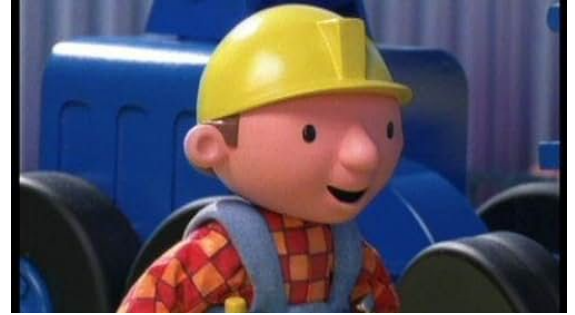
The reviews section for automotive mobility products is relatively active. We may adopt the ability to include pictures and base our review feature off of Amazon's review section. We may also include purchase links to products to sites like Amazon. <https://www.amazon.com>



Reddit r/disability

While we were unable to find a subreddit specifically for automotive mobility, there exist many forums on sites like reddit that allow handicapped persons to share a space to talk about their experiences. Our site would allow those with special needs to discuss, find resources on, and review products related to automotive mobility. <https://www.reddit.com/r/disability>

High Priority Challenges



Immediate Challenges and Design Issues to be tackled first.

1. Create initial product database with rating system.
2. Product submission process.
3. Admin capabilities to quality check submitted reviews by users.
4. User profile page to manage reviews and information shared with website.
5. Search filters for reviews and product types within database.
6. 3D Visualization of test setups using Three.js.

High Priority Challenges

Preventative thoughts for potential threat actors.

1. Limiting the number of reviews an account can post, potential to flood the database with a bunch of bot reviews.
2. Authentication for user registration and password reset ability.